

- 5 Pneumonia is a severe lung disease that can interfere with gas exchange. A person with pneumonia can be connected to an ECMO machine. This machine performs the gas exchange functions of the lungs.

A cannula (tube) is inserted into the right atrium and this takes blood to the ECMO machine. In the ECMO machine, blood is passed firstly to an artificial pump and then to an oxygenator, where gas exchange occurs. The blood is then warmed and returns by another cannula to the vena cava.

- (a) Complete Fig. 5.1 to show how the ECMO machine is connected to the right atrium and to the vena cava. Use a single line to represent each cannula.

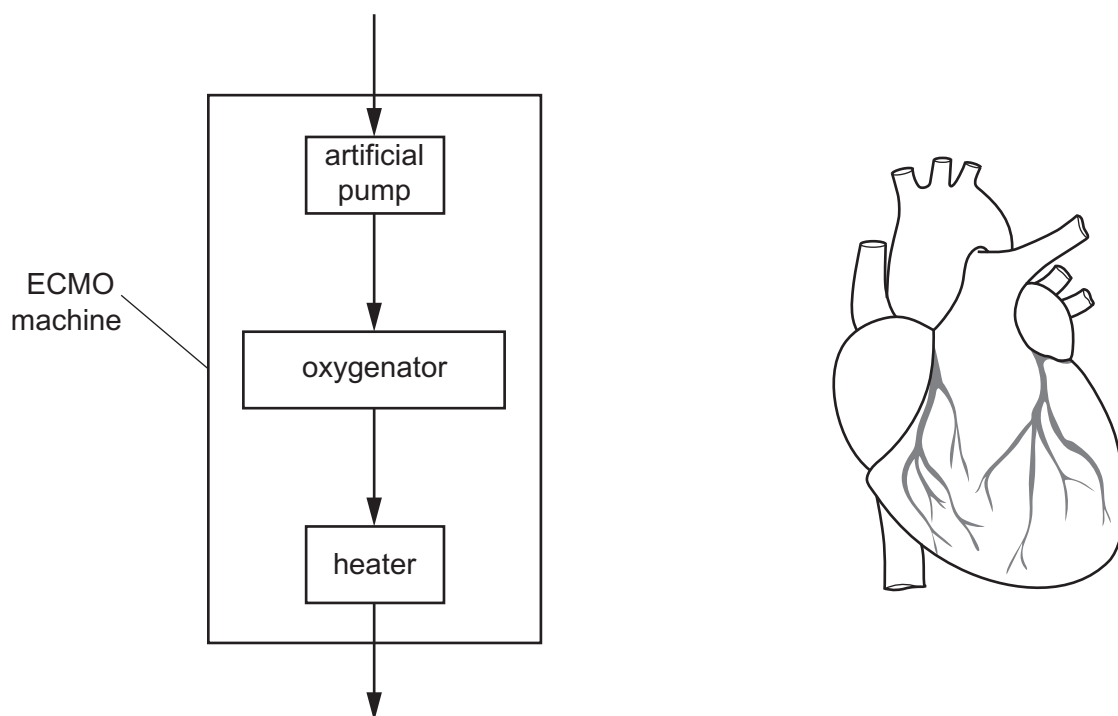


Fig. 5.1

[1]

- (b) In the oxygenator, a partially permeable membrane separates the blood from air that has been enriched with extra oxygen.
- (i) State the name of a structure in the gas exchange system that has the same function as the partially permeable membrane of the oxygenator.

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- (ii) In the oxygenator, blood and oxygen-enriched air flow in opposite directions.

Suggest **and** explain how the oxygenator carries out the functions of gas exchange that normally occur in the lungs.

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- (c) Fig. 5.2 is a photomicrograph showing a transverse section of part of the human aorta.

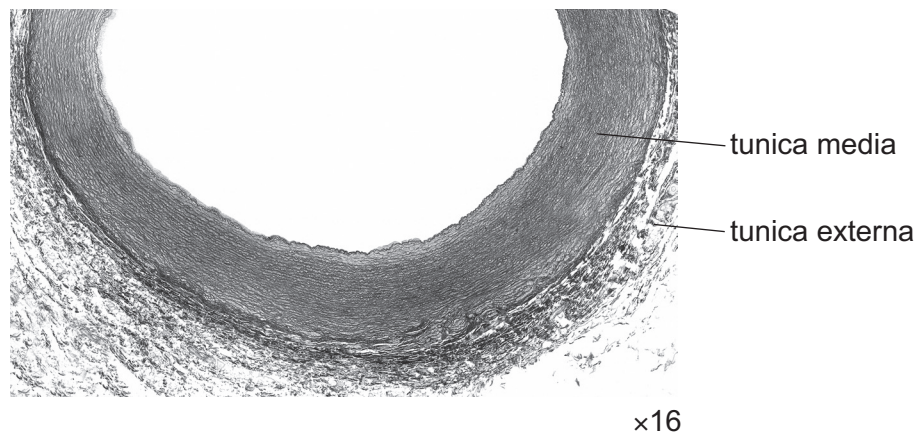


Fig. 5.2

Explain how the structure of the tunica media in Fig. 5.2 is different from the structure of the tunica media in a muscular artery **and** relate the difference to the function of the aorta.

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- (d) Some biologists investigated the transport of carbon dioxide in the blood of *Caiman latirostris*, a type of reptile.

The biologists found that when *C. latirostris* respire:

- most of the carbon dioxide is converted into hydrogencarbonate ions in red blood cells
- the hydrogencarbonate ions combine with haemoglobin inside the red blood cells
- the hydrogencarbonate ions remain combined with haemoglobin until the blood reaches the lungs.

- (i) Explain why the physiology of *C. latirostris* requires carbonic anhydrase.

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- (ii) Explain why the physiology of *C. latirostris* does **not** require the chloride shift.

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[Total: 10]

Question 6 starts on page 18.